
Figure 1: The IBM Bluemix online platform


Figure 2: IBM Bluemix in a Platform as a Service (PaaS) mode


Figure 3: Selecting the IoT application from IBM Bluemix

Table 1: Hypervisors used in the industry

| Hypervisor | Cloud service provider |
|---|---|
| Xen | • Amazon EC2<br>• IBM SoftLayer<br>• Fujitsu Global Cloud Platform<br>• Linode<br>• OrionVM |
| ESXi | VMware Cloud |
| KVM | • Red Hat<br>• HP<br>• Del<br>• Rackspace |
| Hyper-V | Microsoft Azure |

## IBM Bluemix - *bluemix.net*

IBM Bluemix is a leading cloud service provider offering a number of services including Big Data analytics, mobile applications, NoSQL databases, the Internet of Things (IoT), natural language processing, sentiment analysis, and a lot of related high performance computing services.

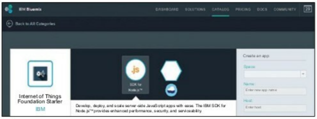


Figure 4: Specifying the app name in IBM Bluemix

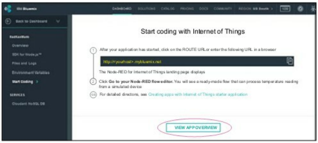


Figure 5: View the app details

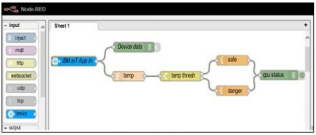


Figure 6: Configuring the IoT app in Node RED

The IBM Bluemix Platform can be accessed using the 'freemium' model, and there is the provision to use boilerplates. The latter simplifies the creation and updation of live code, enabling even beginners to use IBM cloud services without any difficulty.

In IBM Bluemix, there is a detailed catalogue of a number of cloud services including the programming platform for PHP, Java, Ruby, Python and many other languages. Bluemix can be used as a *PaaS* for hosting and running multiple applications.

After logging in, click on the *Catalog* button to view and select different platforms and services provided by IBM Bluemix.

## Creating an Internet of Things (IoT) application in Bluemix

Let's create an IoT application in IBM Bluemix. On the *Catalog* page, click 'Internet of Things Application Foundation Starter'. This is a simulation scenario for sensing the temperature.

As in Figure 4, the app name is to be specified while creating the live cloud application. Using this unique name, the application is identified and executed on the cloud servers.